Algebra & Functions — Timed Simulation 3

IBEW / ETA math section format

33	46	~84 sec	No
questions	minutes	per question	calculator

About this test

- All 33 questions from your photos of Timed Simulation 3, retyped clean and printable, in the original order. Wording, numbers and answer choices are kept as they appeared on screen.
- Graphs are redrawn on a clean grid to match what was on screen.
- Multiple choice, A–D. No calculator. No penalty for guessing — answer everything.

How to take it

- Set a timer for 46 minutes and don't stop it.
- Skip anything that takes more than about 90 seconds and come back to it.
- The answer key with a short explanation for every question is at the back.

What's on it

Simplifying, expanding & factoring	Questions 1, 13, 20, 28
Number series	Questions 5, 15, 27, 29, 33
Plugging in & solving	Questions 4, 7, 8, 11, 12, 14, 19, 23
Isolating a variable	Questions 24, 30, 31
Variable relationships (which statement is true)	Questions 6, 10, 22, 25, 26, 32
Same relationship: tables, graphs, words	Questions 9, 16, 18, 21
Graphing & systems	Questions 2, 3, 17

Question 1

Simplify the polynomial:

$(-3x + 2)(5 - x) + x^2$

A) $3x^2 - 15x + 10$

B) $4x^2 - 17x + 10$

C) $-2x^2 - 15x + 10$

D) $2x^2 - 17x + 10$

WORKSPACE

Question 2

Solve the following system of equations:

i. $x + y = 7$

ii. $x - y = 1$

A) (3, 4)

B) (4, 3)

C) (−4, 3)

D) (−3, 4)

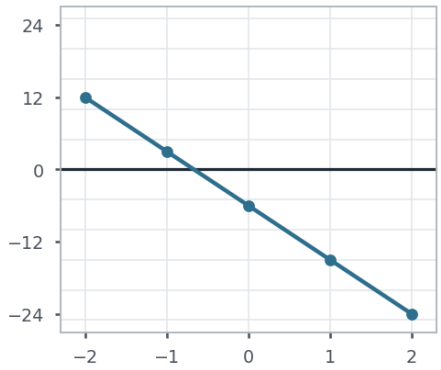
WORKSPACE

Question 3

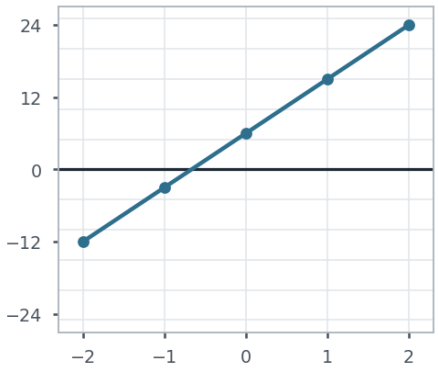
Which of the graphs represents the following table of values?

x	y
−2	12
−1	3
0	−6
1	−15
2	−24

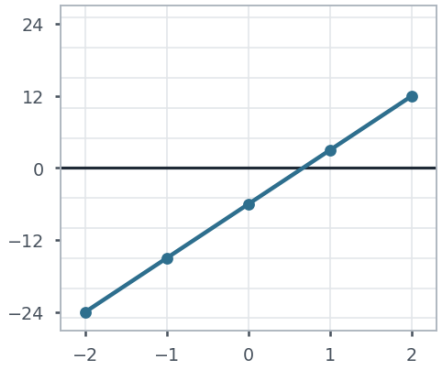
A)



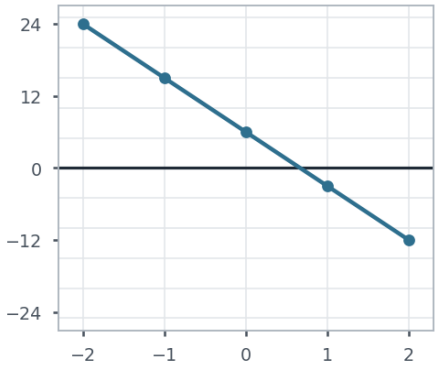
B)



C)



D)



WORKSPACE

Question 4

Consider the following formula:

$2E + 3D = 4DE - 3D$

If $E = 2$, which of the following statements is true for this formula?

A) $D > E$

B) $D = E$

C) $D + E < 0$

D) $D - E < 0$

WORKSPACE

Question 5

What is the next number in the following series of numbers?

7 11 13 17 19 23 ?

A) 24

B) 26

C) 27

D) 29

WORKSPACE

Question 6

Consider the following formula:

$5a = b - 15$

Which of the following statements is true for this formula?

A) When the value of a is less than 3, b is positive.

B) When the value of a is greater than -3, b is positive.

C) When the value of b is greater than -15, a is negative.

D) When the value of b is greater than 15, a is negative.

WORKSPACE

Question 7

Evaluate the value of the unknown variable:

$c = \frac{d}{e}$ and $e = \frac{fg}{h}$

$d = 14$, $f = 4$, $g = 8$, and $h = 16$. Find c.

A) 256

B) 2

C) −28

D) 7

WORKSPACE

Question 8

Consider the following formula:

$$AB = BC + 2(C - 0.5)$$

If $A = 4$ and $B = 5$, what is the value of C?

A) 2

B) 3

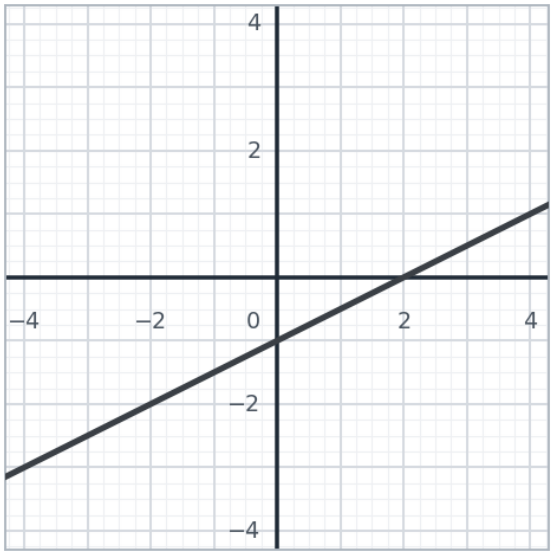
C) 7

D) 10

WORKSPACE

Question 9

Consider the following graph:



Which of the following options represents the same relationship as demonstrated in this graph?

A) $x - 2y = 2$

B)

x	y
−4	−3
−2	−2
0	−1
2	1
4	2

C) A linear graph has a negative slope and intercepts with the axes at y = (−2) and x = 2.

D) None of the options demonstrate the same relationship as the graph.

WORKSPACE

Question 10

Consider the following formula:

$$Q = \frac{R}{2}$$

Which of the following statements is true for this formula?

A) The difference between the values of R and Q diminishes for greater positive values of Q.

B) The difference between the values of R and Q is constant.

C) For positive values, the difference between Q and zero is less than the difference between R and zero.

D) For positive values, the difference between Q and zero is greater than the difference between R and zero.

WORKSPACE

Question 11

Solve the proportion:

$$\frac{6}{2z+3} = \frac{9}{3z+2+z}$$

A) $\frac{1}{2}$

B) $1\frac{1}{2}$

C) $2\frac{1}{2}$

D) $3\frac{1}{2}$

WORKSPACE

Question 12

Evaluate the value of the unknown variable:

$P = 60 \qquad R = 6 \qquad t = 8$

$$a = \frac{R - P}{t}$$

$a = ?$

A) 5.25

B) -6

C) -6.75

D) 7.5

WORKSPACE

Question 13

Simplify the following expression to its simplest form.

Consider $B = 2$ and $C = 6$:

$$A(B(C-4)) + C(A-B)(A+1)$$

A) $6A^2 - 2A - 12$

B) $6A^2 + 10A - 2$

C) $A^2 - 2A - 2$

D) $8A - 1$

WORKSPACE

Question 14

Consider the following formula:

$A(A + 2B + 2C) = 11 + B$

If $A = 2$ and $B = 1$, what is the value of C?

A) 1

B) 2

C) -2

D) -1

WORKSPACE

Question 15

What is the next number in the following series of numbers?

51 44 37 30 23 ?

A) 16

B) 17

C) 20

D) 21

WORKSPACE

Question 16

Consider the following function:

$y = 3x + 1$

Which of the following choices represents the same relationship as the given function?

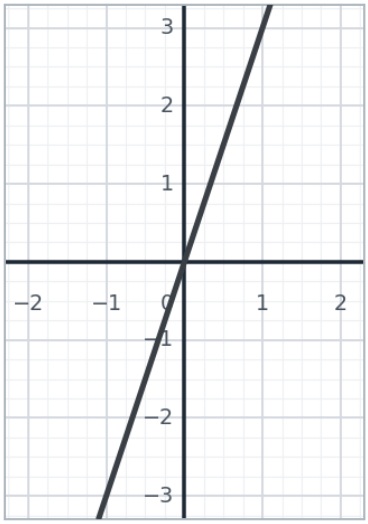
A) The function's graph line intercepts the y-axis at the point y = 1 and the x-axis at the point x = −1/3.

B)

x	y
1	0
4	1
7	2
10	3
13	4

C) $x = \frac{y}{3} + 1$

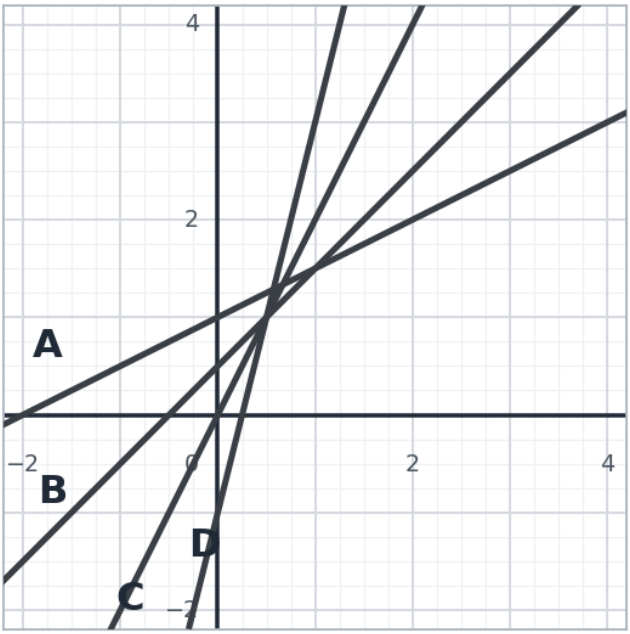
D)



WORKSPACE

Question 17

The lines below depict four linear functions. Which of the functions is depicted by line B?



A) $4x = 2y$

B) $4x - 1 = y$

C) $2x + 1 = 2y$

D) $x + 2 = 2y$

WORKSPACE

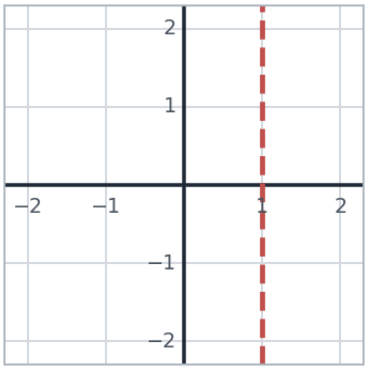
Question 18

Consider the following information:

$y = Ax - b$, where $A = 0$ and $b = 1$

Which of the following options represents the same relationship as the aforementioned information?

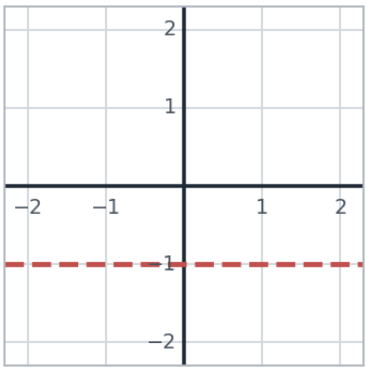
A)



B)

x	y
0	1
−1	1
−2	1
−3	1
−4	1

C)



D)

x	y
−1	0
−1	1
−1	2
−1	3
−1	4

WORKSPACE

Question 19

Consider the following formula:

$A = B + 3(C - 3)$

If $A = 10$ and $B = 1$, what is the value of C?

A) 3

B) 9

C) 4

D) 6

WORKSPACE

Question 20

Factor the polynomial:

$x^2 - 14x + 24$

A) $(x - 4)(x - 6)$

B) $(x - 12)(x - 2)$

C) $(x - 8)(x - 3)$

D) $(x - 10)(x + 24)$

WORKSPACE

Question 21

Consider the following table, which contains values that are found on the graph of the function:

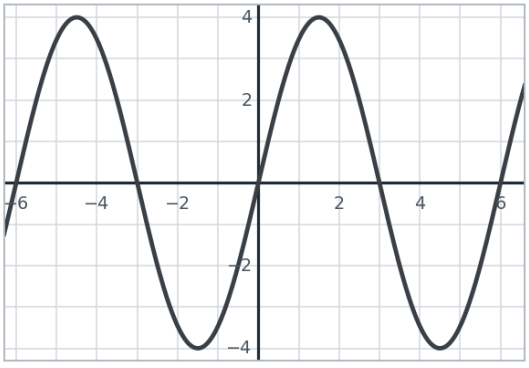
x	y
0	4
1	2
2	0
3	4
4	2
6	0

Which of the following options is accurate given the aforementioned information?

A) $y = 4x - 2$

B) The function is definitely not a linear function.

C) $x = y - x + 2$

D) 

WORKSPACE

Question 22

Consider the following formula:

$2F - G = 0$

Which of the following statements is true for this formula?

A) F is always greater than G.

B) If F is positive, G is negative.

C) If F is positive, G is positive too.

D) F is always lower than G.

WORKSPACE

Question 23

Evaluate the expression:

For $x = 11$,

$x^2 - 15x + 56 = ?$

A) 12

B) 11

C) -56

D) 15

WORKSPACE

Question 24

Isolate the variable from the formula:

$AB + C = D + C$

$B = ?$

A) DA

B) DAC

C) $\frac{D + C}{A}$

D) $\frac{D}{A}$

WORKSPACE

Question 25

Consider the following formula:

$Y = 5 - 2W$

Which of the following statements is true for this formula?

A) If W is less than 10, Y is positive.

B) If W is greater than -2.5, Y is negative.

C) If W is greater than -2.5, Y is positive.

D) If W is less than 2.5, Y is positive.

WORKSPACE

Question 26

Consider the following formula:

$$x = \frac{1}{y}$$

Which of the following statements is true for this formula?

A) When x is positive, $y > 1$.

B) When x is positive, $y < 1$.

C) When x is negative, $y > 1$.

D) When x is negative, $y < 1$.

WORKSPACE

Question 27

What is the next number in the following series of numbers?

7 25 43 61 ?

A) 73

B) 79

C) 80

D) 83

WORKSPACE

Question 28

Consider the following formula:

$y = x(x + 8)(3x^2 + 1)$

Which of the following formulas is equivalent to this one?

A) $3x^3 + 24x^2 + x + 8$

B) $3x^4 + 24x^2 + x + 8$

C) $3x^4 + 25x^2 + 8$

D) $3x^4 + 24x^3 + x^2 + 8x$

WORKSPACE

Question 29

What is the next number in the following series of numbers?

3 5 9 17 33 ?

A) 45

B) 57

C) 65

D) 71

WORKSPACE

Question 30

Isolate the variable from the formula:

$XY - ZA = BR$

$A = ?$

A) $\frac{BRXY}{-Z}$

B) $\frac{BR - XY}{-Z}$

C) $\frac{BR}{XYZ}$

D) $\frac{BR + XY}{Z}$

WORKSPACE

Question 31

Isolate the variable from the formula:

$$F = \frac{Gmn}{r}$$

$G = ?$

A) $\frac{mn}{rF}$

B) $\frac{rF}{mn}$

C) $\frac{Fmn}{r}$

D) $\frac{rmn}{F}$

WORKSPACE

Question 32

Consider the following formula:

$$\frac{T}{5} = \frac{S}{3} + 2$$

Which of the following statements is true for this formula?

A) If S is less than -3, T is positive.

B) If S is less than -6, T is positive.

C) If S is less than -3, T is negative.

D) If S is greater than -3, T is positive.

WORKSPACE

Question 33

What is the next number in the following series of numbers?

0.5 2 1 2 2 4 8 ?

A) 32

B) 16

C) 8

D) 64

WORKSPACE

Answer Key

1. B	**10. C**	**19. D**	**28. D**
2. B	**11. C**	**20. B**	**29. C**
3. A	**12. C**	**21. B**	**30. B**
4. B	**13. A**	**22. C**	**31. B**
5. D	**14. A**	**23. A**	**32. D**
6. B	**15. A**	**24. D**	**33. A**
7. D	**16. A**	**25. D**	
8. B	**17. C**	**26. D**	
9. A	**18. C**	**27. B**	

1. B FOIL: $-15x + 3x^2 + 10 - 2x = 3x^2 - 17x + 10$. Add x^2: $4x^2 - 17x + 10$

2. B Add the equations: $2x = 8 \rightarrow x = 4$. Then $y = 7 - 4 = 3$.

3. A Starts at 12, ends at -24, and drops 9 each step: $y = -9x - 6$.
Graph A falls from 12 to -24. D also falls but runs from 24 down to -12.

4. B $4 + 3D = 8D - 3D = 5D \rightarrow 4 = 2D \rightarrow D = 2$. So $D = E$.

5. D These are the prime numbers in order. The next prime after 23 is 29.

6. B $b = 5a + 15$, which is 0 at $a = -3$. Above -3, b is positive ✓
A fails at $a = -5$ ($b = -10$). D: $b = 20$ gives $a = 1$, positive.

7. D $e = \frac{4 \times 8}{16} = 2$. $c = \frac{14}{2} = 7$

8. B $20 = 5C + 2C - 1 \rightarrow 21 = 7C \rightarrow C = 3$

9. A The line crosses at (0, -1) and (2, 0): $y = \frac{1}{2}x - 1$.
Times 2: $2y = x - 2 \rightarrow x - 2y = 2$ ✓. The table breaks at $x = 2$ (should be 0).

10. C Q is half of R, so for positive values Q is always closer to zero than R is.
$R - Q = \frac{R}{2}$, which grows (so A and B are false).

11. C Cross-multiply: $6(4z + 2) = 9(2z + 3) \rightarrow 24z + 12 = 18z + 27 \rightarrow 6z = 15 \rightarrow z = 2\frac{1}{2}$

12. C $\frac{6 - 60}{8} = \frac{-54}{8} = -6.75$

Answer Key (cont.)

13. A $A(2 \times 2) + 6(A-2)(A+1) = 4A + 6(A^2 - A - 2) = 6A^2 - 2A - 12$

14. A $2(4 + 2C) = 12 \;\rightarrow\; 4 + 2C = 6 \;\rightarrow\; C = 1$

15. A Subtract 7 each time: $23 - 7 = 16$

16. A y-intercept: $x = 0$ gives $y = 1$ ✓. x-intercept: $0 = 3x + 1 \;\rightarrow\; x = -\frac{1}{3}$ ✓

The table is backwards ($x = 3y + 1$). C works out to $y = 3x - 3$. The graph is $y = 3x$.

17. C Line B rises 1 for every 1 right and crosses the y-axis at $\frac{1}{2}$: $y = x + \frac{1}{2}$.

Times 2: $2y = 2x + 1$ ✓. (A is $x + 2 = 2y$, C is $4x = 2y$, D is $4x - 1 = y$.)

18. C $y = 0 \cdot x - 1 = -1$: a flat line at $y = -1$. That's graph C.

Trap: table B is $y = +1$, the wrong sign.

19. D $10 = 1 + 3(C - 3) \;\rightarrow\; 9 = 3(C - 3) \;\rightarrow\; 3 = C - 3 \;\rightarrow\; C = 6$

20. B Two numbers that multiply to 24 and add to -14: -12 and -2.

21. B It drops 2, drops 2, then jumps up 4 — a straight line can't change direction, so it's not linear.

The wave graph passes through (0, 0), not (0, 4).

22. C $G = 2F$, so G always has the same sign as F. (A and D fail because it flips for negatives.)

23. A $121 - 165 + 56 = 12$

24. D Subtract C from both sides: $AB = D$. Divide by A: $B = \frac{D}{A}$

25. D $Y = 0$ at $W = 2.5$. The -2 means Y falls as W rises, so Y is positive below 2.5 ✓

A fails at $W = 5$ ($Y = -5$).

26. D $y = \frac{1}{x}$. If x is negative, y is negative too — so it's always less than 1 ✓

A and B can both fail: $x = \frac{1}{2}$ gives $y = 2$; $x = 2$ gives $y = \frac{1}{2}$.

27. B Add 18 each time: $61 + 18 = 79$

28. D $x(x + 8) = x^2 + 8x$. Times $(3x^2 + 1)$: $3x^4 + x^2 + 24x^3 + 8x$

Answer Key (cont.)

29. C Differences 2, 4, 8, 16 double each time. Next is 32: $33 + 32 = 65$

30. B Subtract XY: $-ZA = BR - XY$. Divide by $-Z$: $A = \frac{BR - XY}{-Z}$

31. B Multiply by r: $rF = Gmn$. Divide by mn: $G = \frac{rF}{mn}$

32. D $T = 0$ when $\frac{S}{3} = -2$, so $S = -6$. Above -6, T is positive — so above -3 it's positive too ✓

A and C both fail somewhere between -6 and -3.

33. A Each number is the two before it multiplied: $0.5 \times 2 = 1$, $2 \times 1 = 2$, $1 \times 2 = 2$, $2 \times 2 = 4$, $2 \times 4 = 8$

Next: $4 \times 8 = 32$